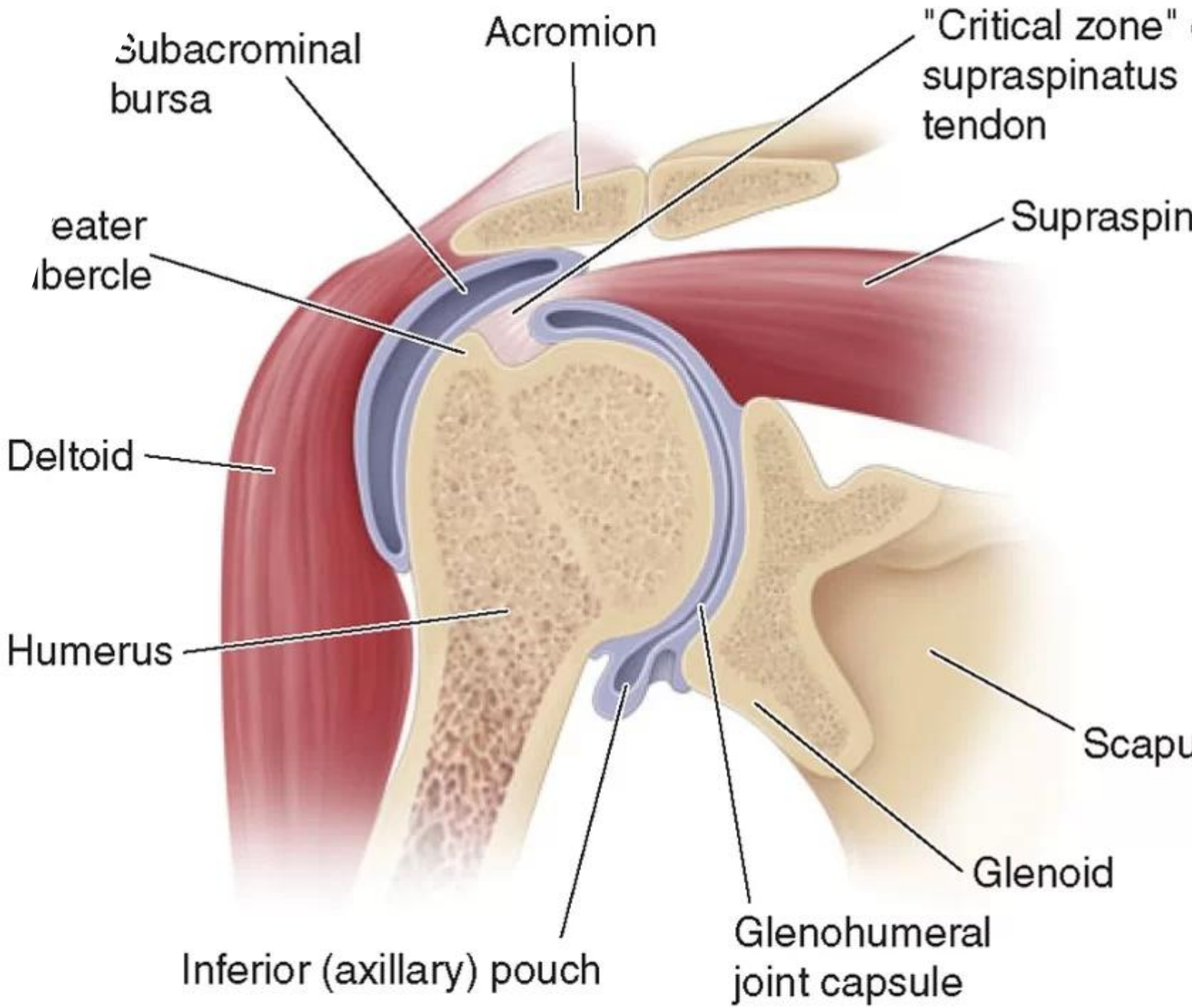
Shoulder Joint Assessment and treatment

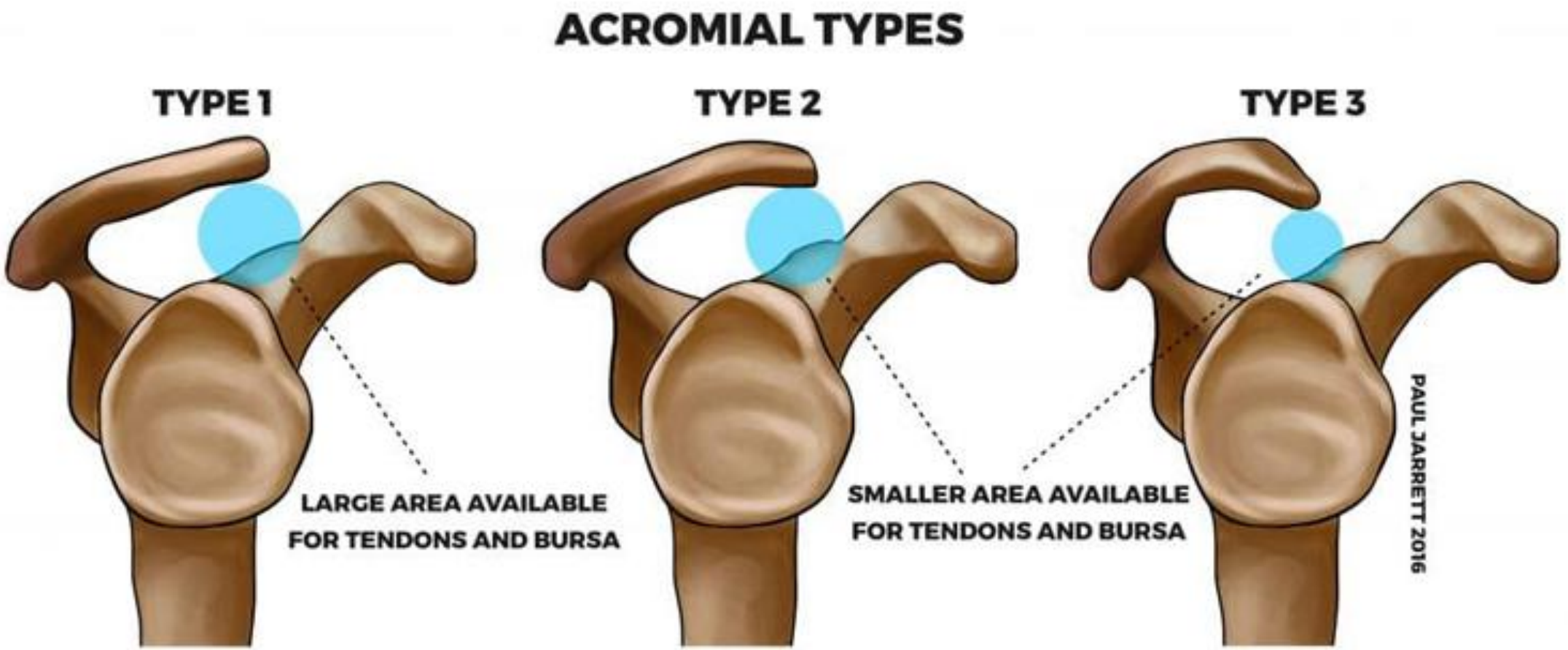
- **Walaa Abu-Taleb**

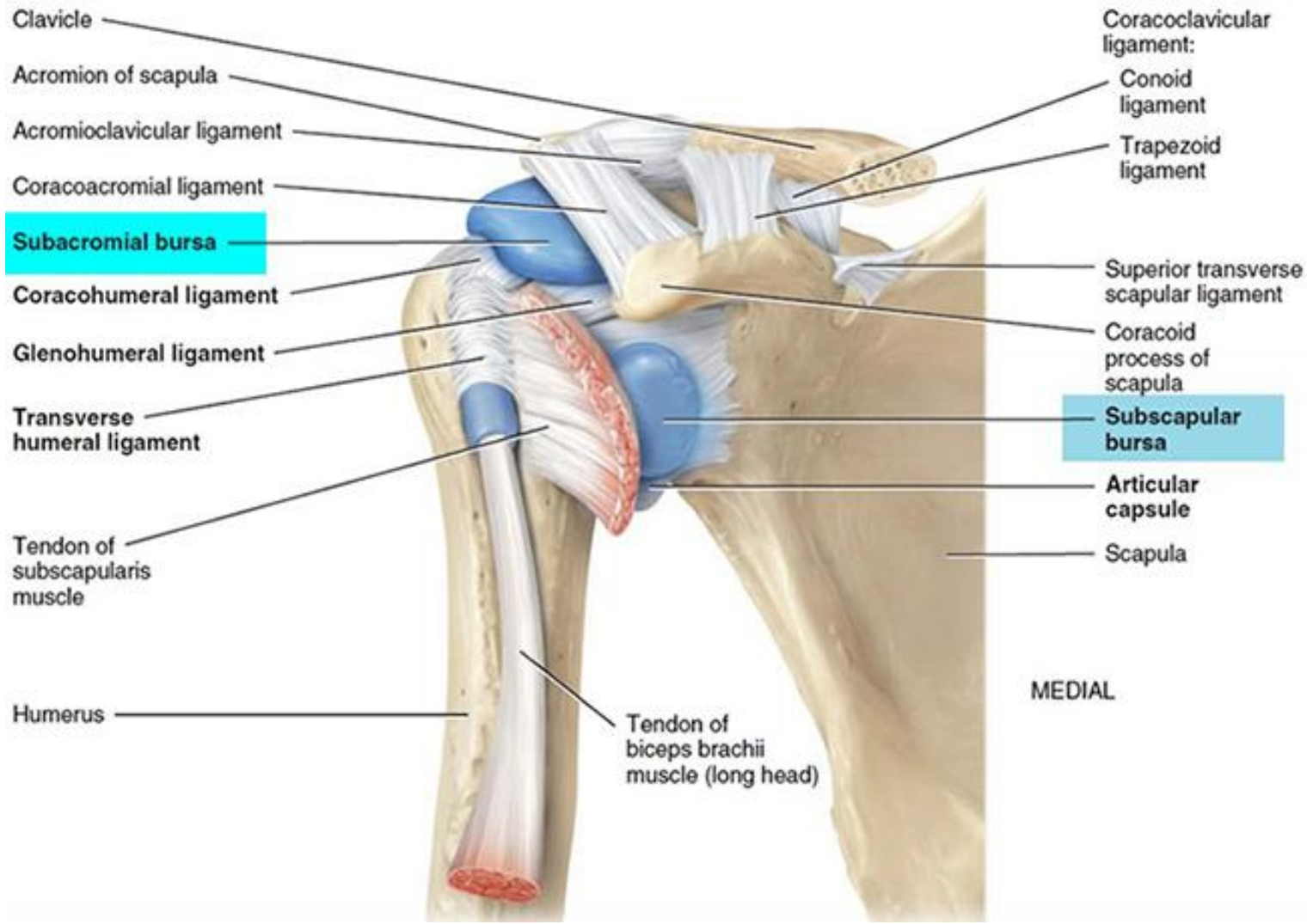


Learning Outcomes

- **By the End of this lesson, you should be able to:**
- Practice MSK assessment relevant to the shoulder joint.




ACROMIAL TYPES
TYPE 1
TYPE 2
TYPE 3
LARGE AREA AVAILABLE
FOR TENDONS AND BURSA
SMALLER AREA AVAILABLE
FOR TENDONS AND BURSA
PAUL JARRETT 2016


Clavicle
Acromion of scapula
Acromioclavicular ligament
Coracoacromial ligament
Subacromial bursa
Coracohumeral ligament
Glenohumeral ligament
Transverse humeral ligament
Tendon of subscapularis muscle
Humerus
Coracoclavicular ligament:
Conoid ligament
Trapezoid ligament
Superior transverse scapular ligament
Coracoid process of scapula
Subscapular bursa
Articular capsule
Scapula
Tendon of biceps brachii muscle (long head)
MEDIAL

Subjective Assessment specific questions

What is the patient's age?	What is the extent and behavior of the patient's pain?
Does the patient support the upper limb in a protected position or hesitate to move it?	Are there any activities that cause or increase the pain?
If there was an injury, what exactly was the mechanism of injury?	Do any positions relieve the pain?
Are there any movements or positions that cause the patient pain or symptoms?	What is the patient unable to do functionally?
Is there any indication of nerve injury?	Does the patient perform a job that requires manual work or overhead activities?
Which hand is dominant?	

Shoulder joint Red flags

Fracture

Dislocation

Muscle / tendon Tear

(Rotator cuff, bicipital tendon)

Ligamentous Labral tear

Cancer

Heart Attack

Stroke

Objective Assessment

Shoulder Joint assessment (Observation)

Types:

A. Informal: dynamic and static posture and movement, quality of movement…etc

B. Formal: observation of the cervical, thoracic spines and upper limbs

1. Observation of Posture.
2. Observation of Muscle
3. Observation of soft tissue
4. Observation of Balance
5. Observation of Gait

Observation

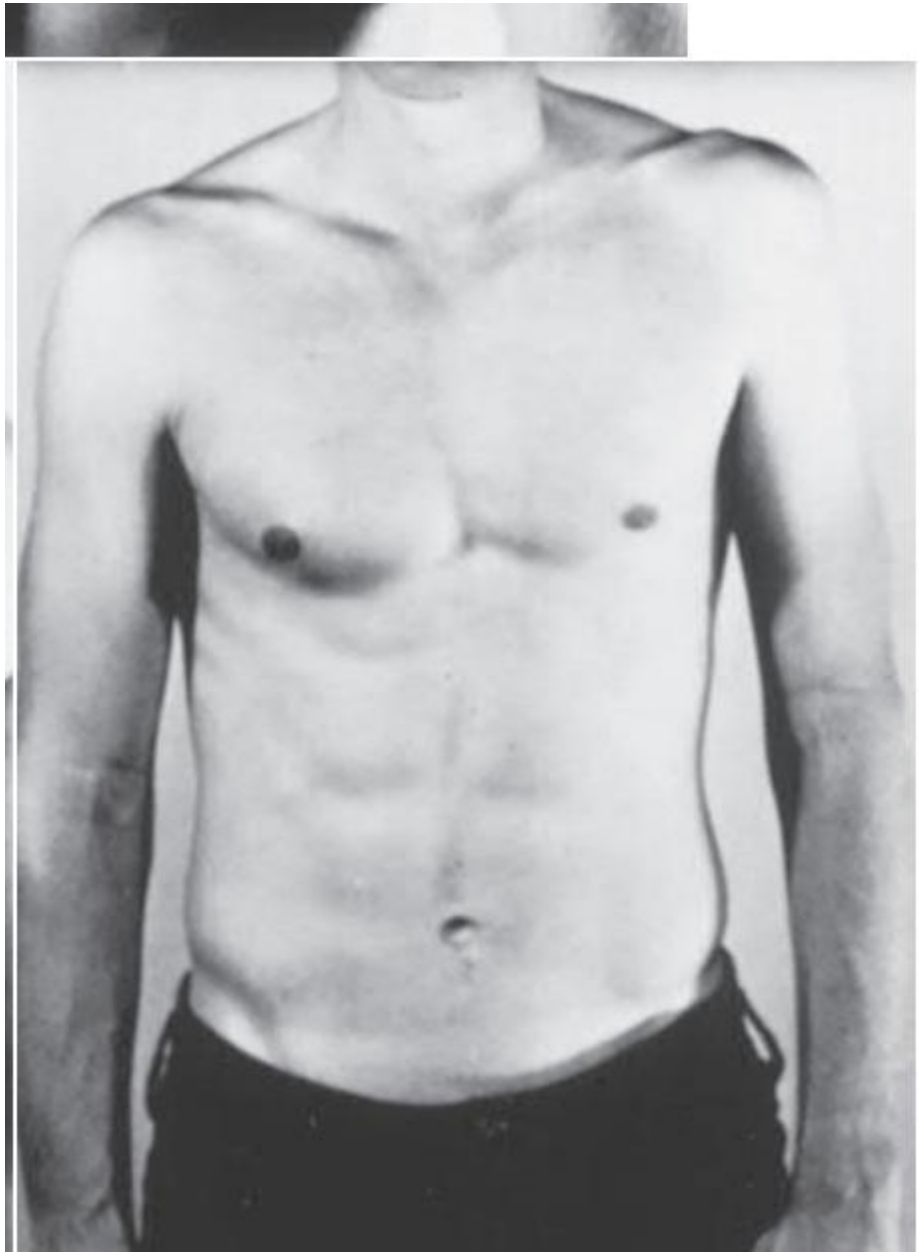
Observe the bony and soft-tissue contours of both shoulders and determine whether they are normal and symmetric.

looks at the head, the cervical spine, the thorax (especially the posterior aspect), and the entire upper limb.

It is important to observe the patient as he or she removes clothes from the upper body and later replaces them.

1- Observation

- Shoulder level asymmetry
- Bones, bony prominent, muscles size,
- May be caused by: muscle imbalance, scoliosis, bad posture, and injuries



Anterior View

- Ensure that the head and neck are in the midline of the body and observing their relation to the shoulders.
- step deformity.
- fountain sign
- sulcus deformity

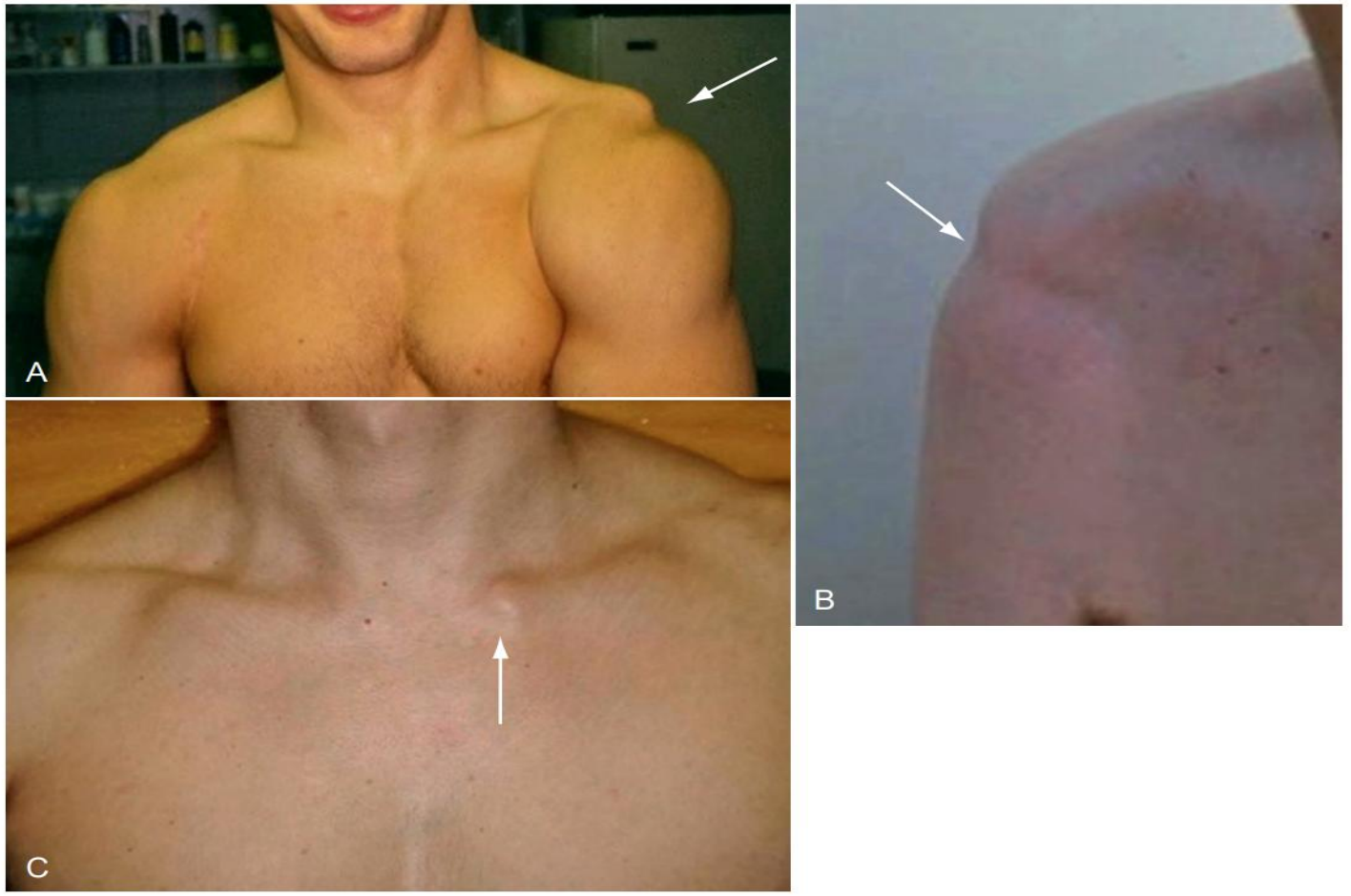
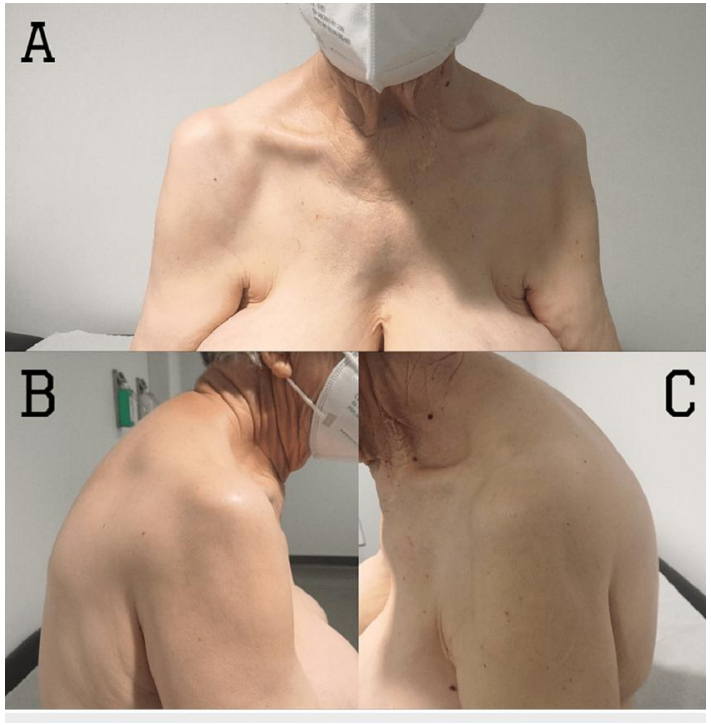


Fig. 5.19 (A) Step deformity resulting from acromioclavicular dislocation *(arrow)*. (B) Subluxation of glenohumeral joint following a stroke (paralysis of deltoid muscle). (C) Anterior dislocation at the sternoclavicular joint *(arrow)*.

Anterior Shoulder dislocation









Observation

- Shoulder dislocation

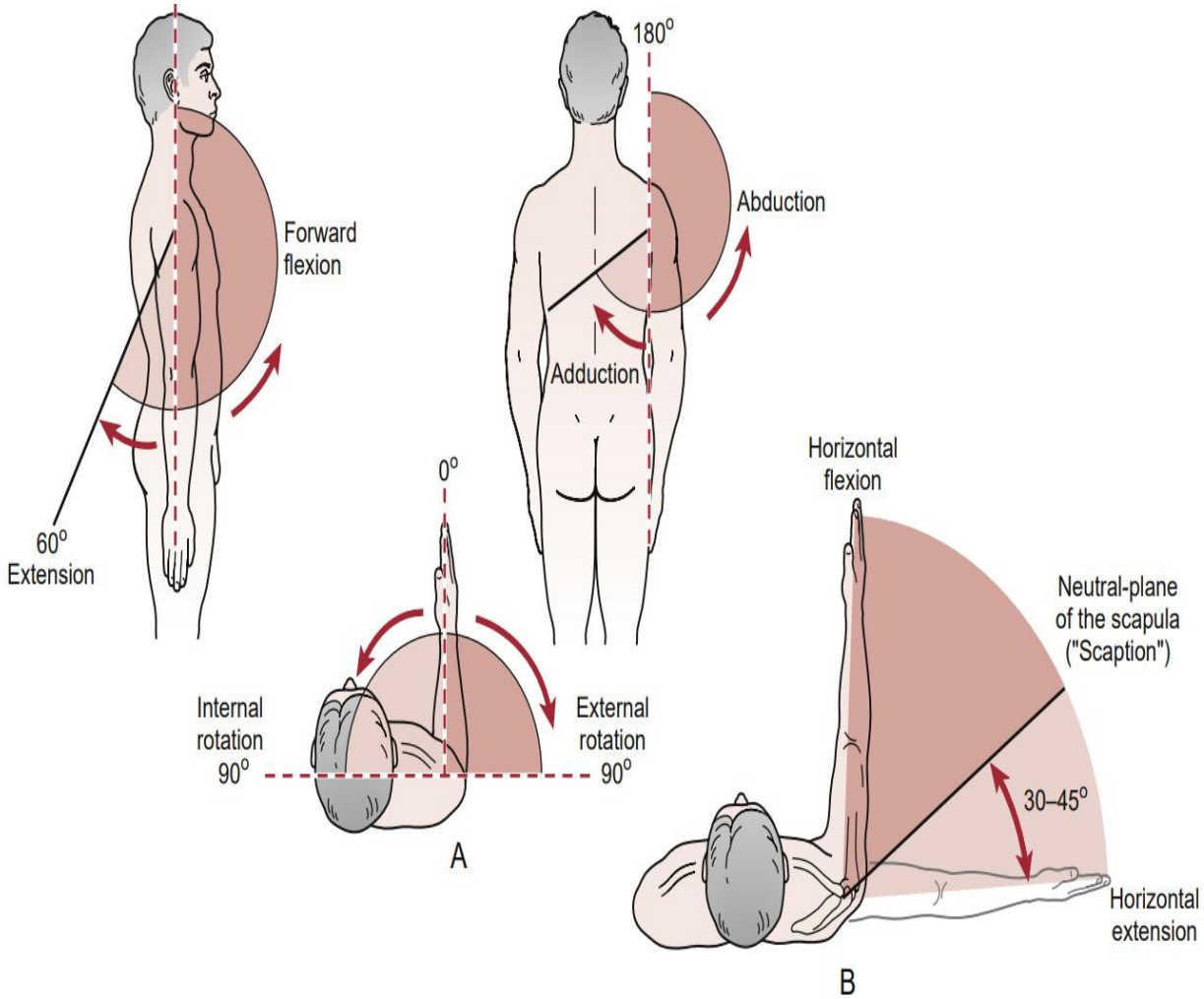
Posterior Observation: Scapular winging

Serratus anterior weakness or long thoracic nerve injury

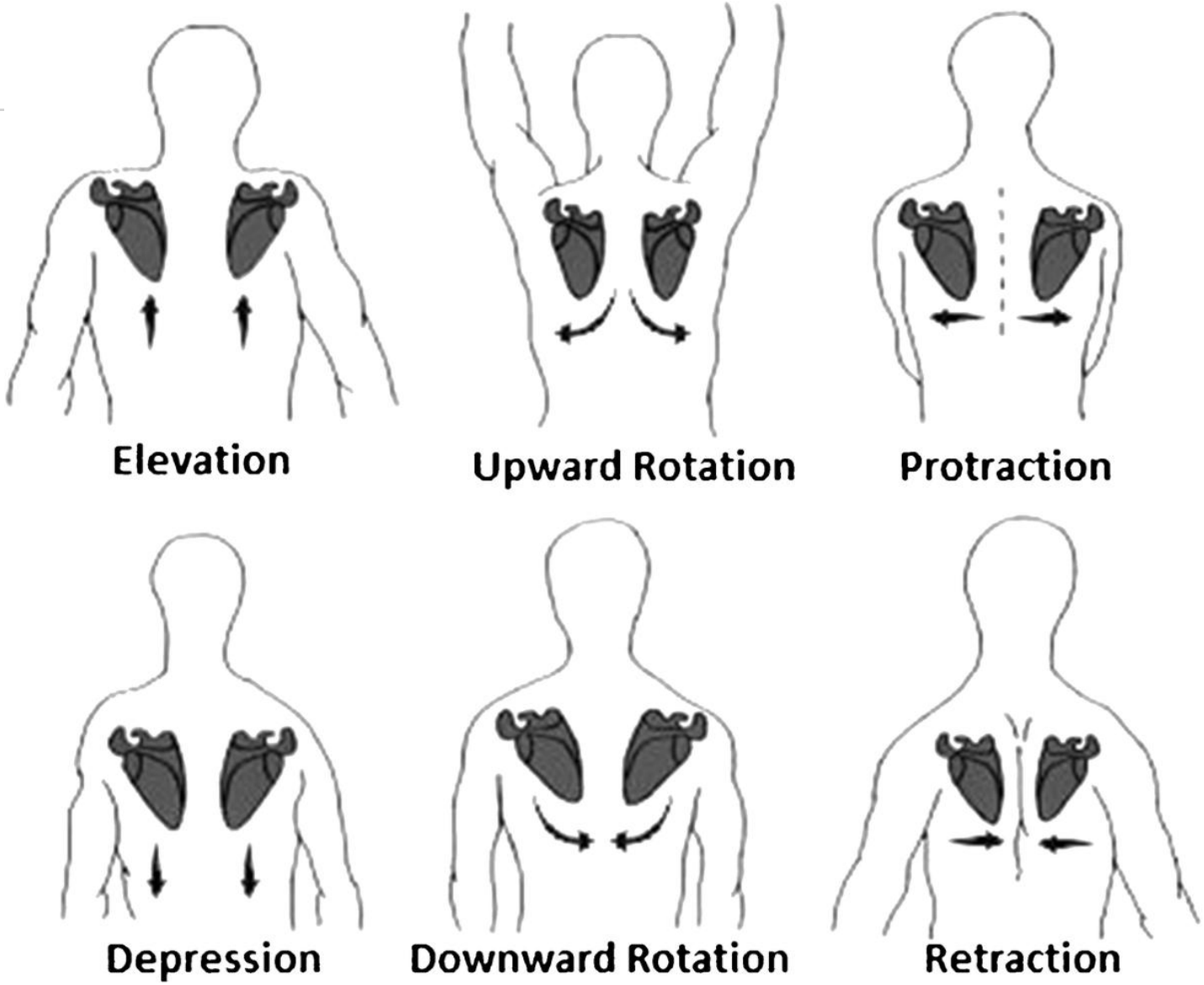


- Rom testing

- What are the movements to assess?
  - Active ROM testing: using Goniometer Flexion/Extension
  - Passive ROM testing : over-pressure and endfeel?
  - Accessory movement testing?
- Do you need to assess it?
- How will we use it?
- What are you looking for?



Scapular Motion



Scapulohumeral Rhythm

Scapulohumeral Rhythm

Phase 1:	Humerus	30° abduction
	Scapula	Minimal movement (setting phase)
	Clavicle	0°–5° elevation
Phase 2:	Humerus	40° abduction
	Scapula	20° rotation, minimal protraction or elevation, and possibly posteriorly tilt
	Clavicle	15° elevation and posterior rotation at sternoclavicular joint
Phase 3:	Humerus	60° abduction, 90° lateral rotation
	Scapula	30° rotation
	Clavicle	30°–50° posterior rotation, up to 15° elevation

Shoulder ROM Ax Goniometer

Direction	Fulcrum/ Axis	Fixed arm	Moving arm
Flexion/ Extension	at the midpoint of the posterior aspect of the glenohumeral joint	parallel to the trunk	parallel to the longitudinal axis of the humerus.
Abduction/ Adduction	at the midpoint of the posterior aspect of the glenohumeral joint	parallel to the trunk	parallel to the longitudinal axis of the humerus.
External/ internal rotation	Olecranon	Vertical to the ground	parallel to the longitudinal axis of the ulna.
Elevation/ Depression	Sternoclavicular joint	Parallel to the sternum	Parallel to the clavicle

Observation

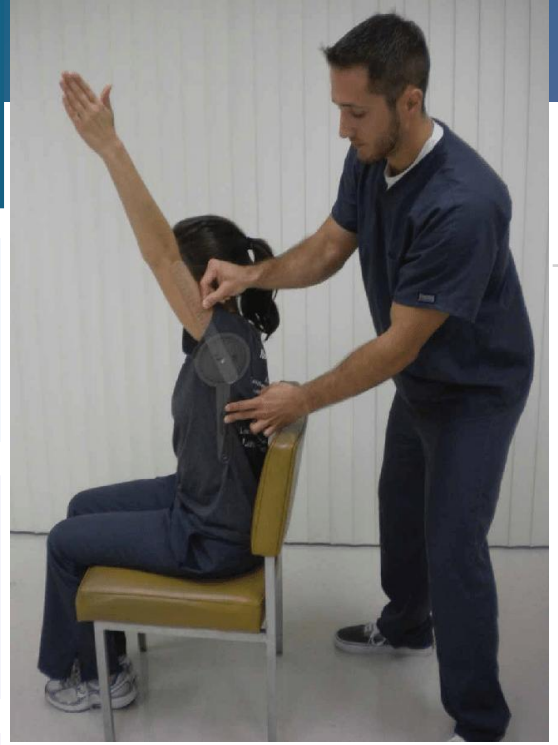
Palpation

ROM

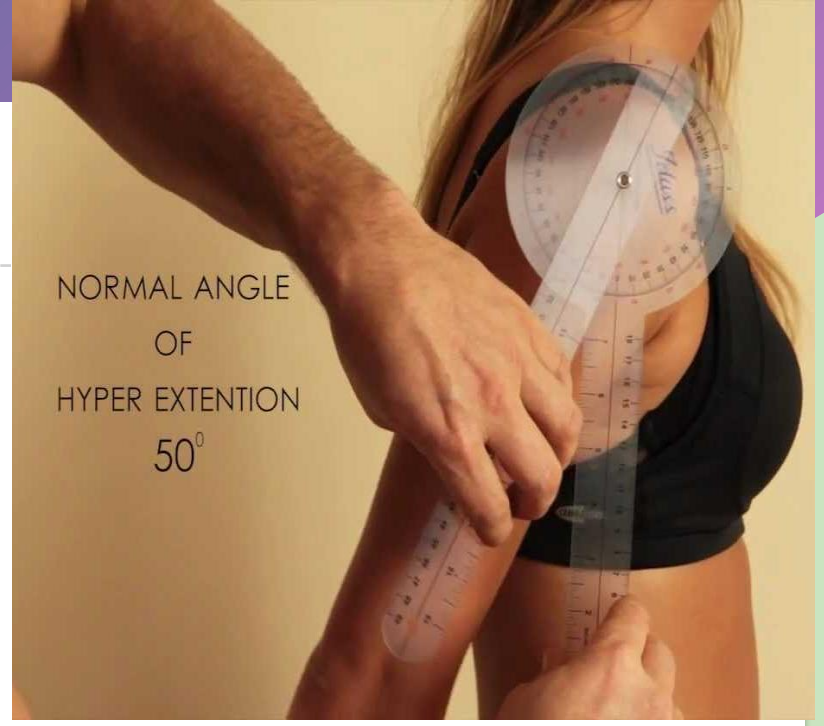
MRC

Special Tests

Neural and vascular test



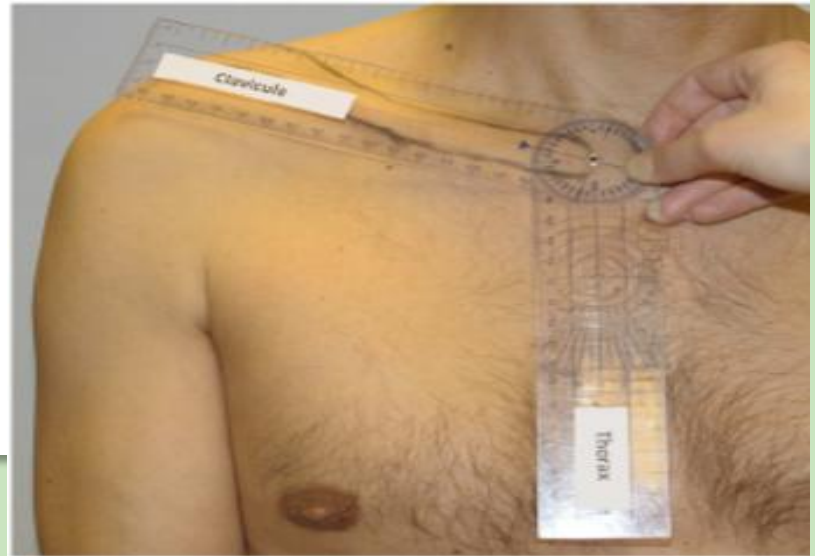




ROM measurements







Observation

Palpation

ROM

MRC

Special Tests

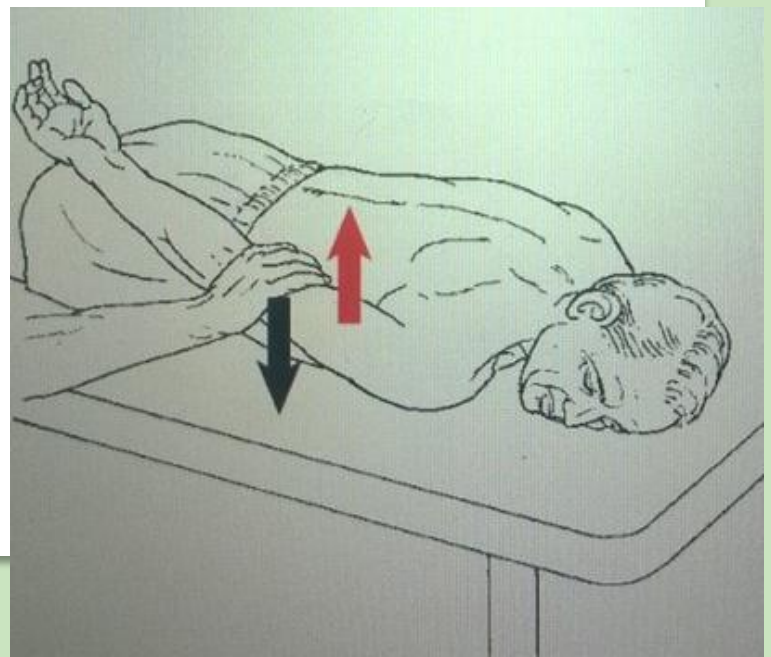
Neural and vascular test

Muscle Tests









Muscles About the Shoulder: Their Actions, Nerve Supply, and Nerve Root Derivation

Action	Acting Muscles	Nerve Supply	Nerve Root Derivation Retraction
Forward flexion	1. Deltoid (anterior fibers)	Axillary (circumflex)	C5, C6 (posterior cord)
	2. Pectoralis major (clavicular fibers)	Lateral pectoral	C5, C6 (lateral cord)
	3. Coracobrachialis	Musculocutaneous	C5–C7 (lateral cord)
	4. Biceps (when strong contraction required)	Musculocutaneous	C5–C7 (lateral cord)
Extension	1. Deltoid (posterior fibers)	Axillary (circumflex)	C5, C6 (posterior cord)
	2. Teres major	Subscapular	CS, C6 (posterior cord)
	3. Teres minor	Axillary (circumflex)	C5, C6 (posterior cord)
	4. Latissimus dorsi	Thoracodorsal	C6–C8 (posterior cord)
	5. Pectoralis major (sternocostal fibers)	Lateral pectoral	C5, C6 (lateral cord)
		Medial pectoral	C8, T1 (medial cord)
	6. Triceps (long head)	Radial	C5–C8, T1 (posterior cord)
Horizontal adduction	1. Pectoralis major	Lateral pectoral	C5, C6 (lateral cord)
	2. Deltoid (anterior fibers)	Axillary (circumflex)	C5, C6 (posterior cord)
Horizontal abduction	1. Deltoid (posterior fibers)	Axillary (circumflex)	C5, C6 (posterior cord)
	2. Teres major	Subscapular	C5, C6 (posterior cord)
	3. Teres minor	Axillary (circumflex)	C5, C6 (brachial plexus trunk)
	4. Infraspinatus	Suprascapular	C5, C6 (brachial plexus trunk)
Abduction	1. Deltoid	Axillary (circumflex)	C5, C6 (posterior cord)
	2. Supraspinatus	Suprascapular	C5, C6 (brachial plexus trunk)
	3. Infraspinatus	Suprascapular	C5, C6 (brachial plexus trunk)
	4. Subscapularis	Subscapular	C5, C6 (posterior cord)
	5. Teres minor	Axillary (circumflex)	C5, C6 (posterior cord)
	6. Long head of biceps (if arm laterally rotated first, trick movement)	Musculocutaneous	C5–C7 (lateral cord)

Adduction	1. Pectoralis major	Lateral pectoral	C5, C6 (lateral cord)
	2. Latissimus dorsi	Thoracodorsal	C6–C8 (posterior cord)
	3. Teres major	Subscapular	C5, C6 (posterior cord)
	4. Subscapularis	Subscapular	C5, C6 (posterior cord)
	5. Coracobrachialis	Musculocutaneous	C5–C7 (lateral cord)
Medial rotation	1. Pectoralis major	Lateral pectoral	C5, C6 (lateral cord)
	2. Deltoid (anterior fibers)	Axillary (circumflex)	C5, C6 (posterior cord)
	3. Latissimus dorsi	Thoracodorsal	C6–C8 (posterior cord)
	4. Teres major	Subscapular	C5, C6 (posterior cord)
	5. Subscapularis (when arm is by side)	Subscapular	CS, C6 (posterior cord)
Lateral rotation	1. Infraspinatus	Suprascapular	C5, C6 (brachial plexus trunk)
	2. Deltoid (posterior fibers)	Axillary (circumflex)	C5, C6 (posterior cord)
	3. Teres minor	Axillary (circumflex)	C5, C6 (posterior cord)
Elevation of scapula	1. Trapezius (upper fibers)	Accessory	CN XI
		C3, C4 nerve roots	C3, C4
	2. Levator scapulae	C3, C4 nerve roots	C3, C4
		Dorsal scapular	C5
	3. Rhomboid major	Dorsal scapular	(C4), C5
	4. Rhomboid minor	Dorsal scapular	(C4), C5
Depression of scapula	1. Serratus anterior	Long thoracic	C5, C6 (C7)
	2. Pectoralis major	Lateral pectoral	C5, C6 (lateral cord)
	3. Pectoralis minor	Medial pectoral	C8, T1 (medial cord)
	4. Lattissimus dorsi	Thoracodorsal	C6–C8 (posterior cord)
	5. Trapezius (lower fibers)	Accessory	CN XI
		C3, C4 nerve roots	C3, C4

Continued

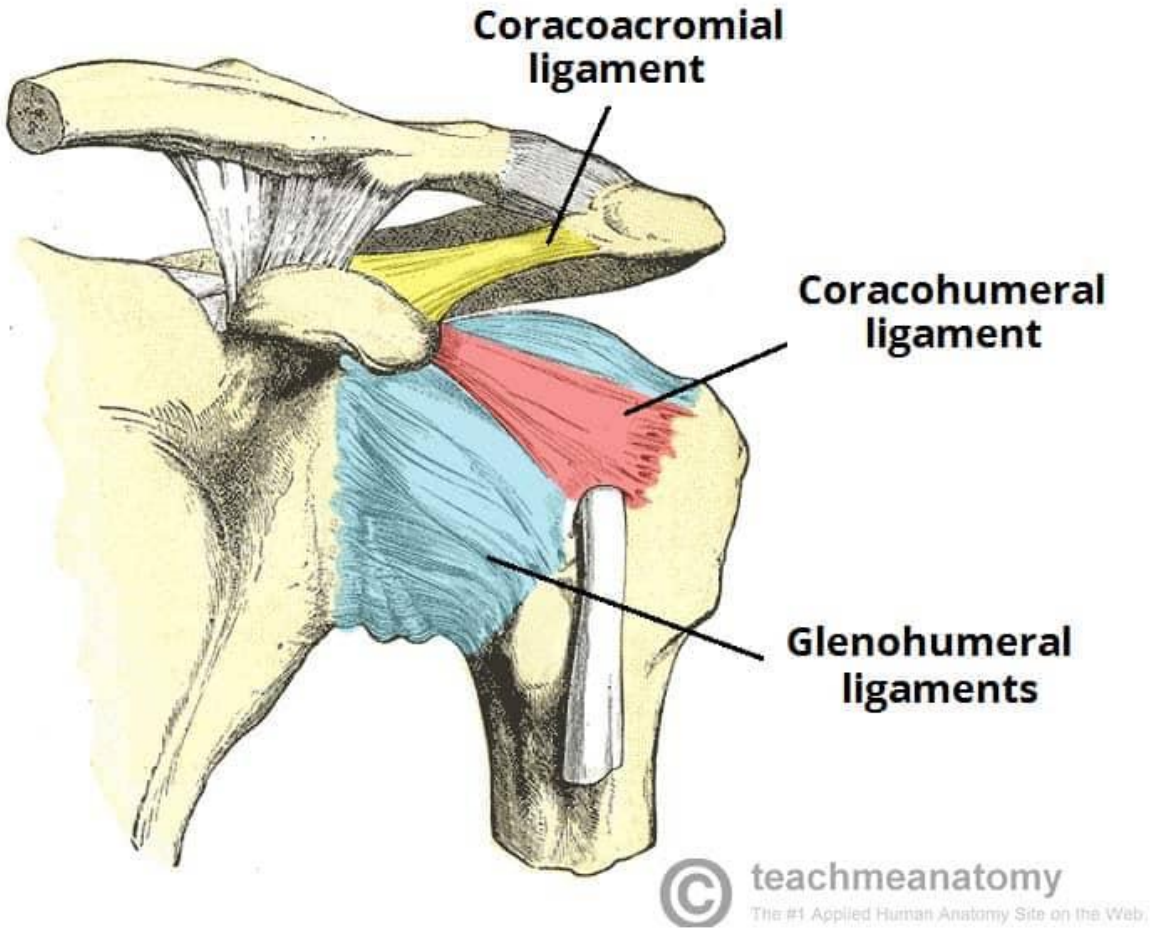
TABLE **5.9**

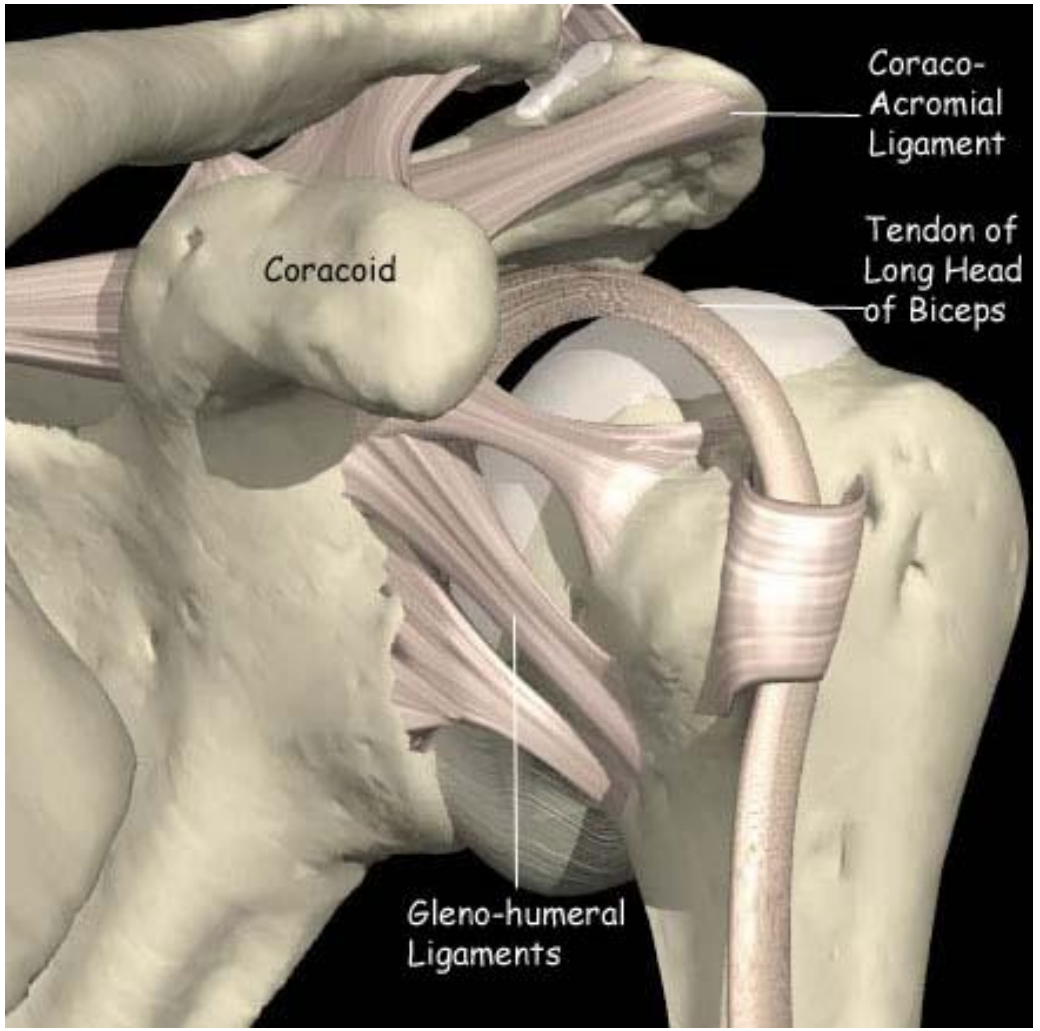
Force Couples About the Shoulder

Movement	Agonist/Stabilizer	Antagonist/Stabilizer
Protraction (scapula)	Serratus anterior[a] Pectoralis major[b] and minor[b]	Trapezius Rhomboids
Retraction (scapula)	Trapezius Rhomboids	Serratus anterior[a] Pectoralis major[b] and minor[b]
Elevation (scapula)	Upper trapezius[b] Levator scapulae[b]	Serratus anterior[a] Lower trapezius[a]
Depression (scapula)	Serratus anterior[a] Lower trapezius[a]	Upper trapezius[b] Levator scapulae[b]
Lateral rotation (upward rotation of inferior angle of scapula)	Trapezius (upper[b] and lower[a] fibers) Serratus anterior[a]	Levator scapulae[b] Rhomboids Pectoralis minor[b]
Medial rotation (downward rotation of inferior angle of scapula)	Levator scapulae[b] Rhomboids Pectoralis minor[b]	Trapezius (upper[b] and lower[a] fibers) Serratus anterior[a]
Scapular stabilization	Upper trapezius[b] Lower trapezius[a] Rhomboids	Serratus anterior[a]
Abduction (humerus)	Deltoid	Supraspinatus
Medial rotation (humerus)	Subscapularis[b] Pectoralis major[b] Latissimus dorsi Anterior deltoid Teres major	Infraspinatus[a] Teres minor Posterior deltoid
Lateral rotation (humerus)	Infraspinatus Teres minor Posterior deltoid	Subscapularis[b] Pectoralis major[b] Latissimus dorsi Anterior deltoid

[a]Muscles prone to weakness.
[b]Muscles prone to tightness.

Joint integrity tests
Collateral Ligaments (Stress tests)





Observation

Palpation

ROM

MRC

Special Tests

Neural and vascular test

Special Tests

Observation

Palpation

Tests

Neural and vascular test

Test of Shoulder integrity (instability)

- Anterior Instability: Apprehension test
- Posterior instability: load and Shift test.
- Inferior instability: Sulcus test

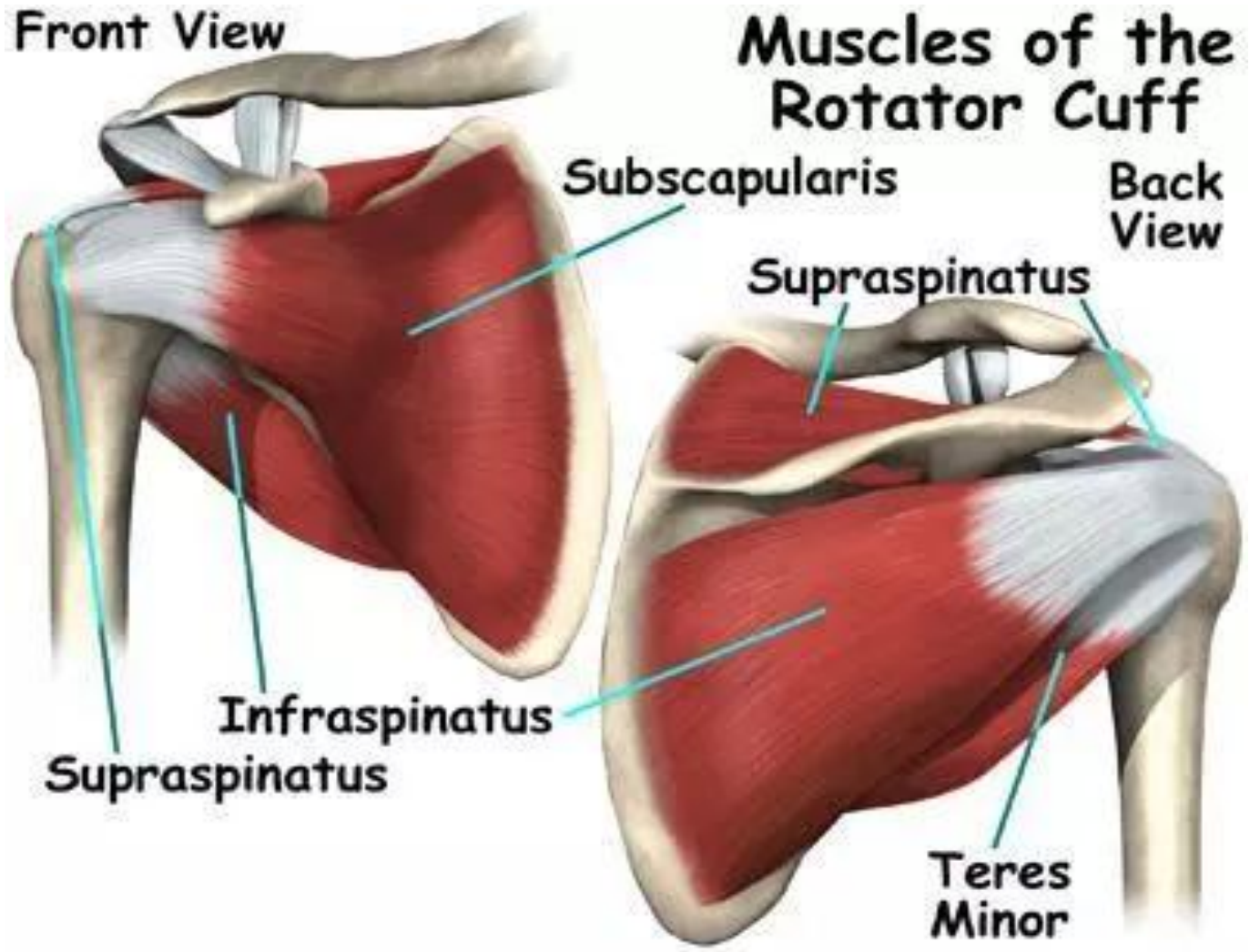






Rotator cuff pathology

Rotator cuff muscles



Rotator Cuff Pathology

Neer's test



Hawkins Kennedy Test



Observation

Palpation

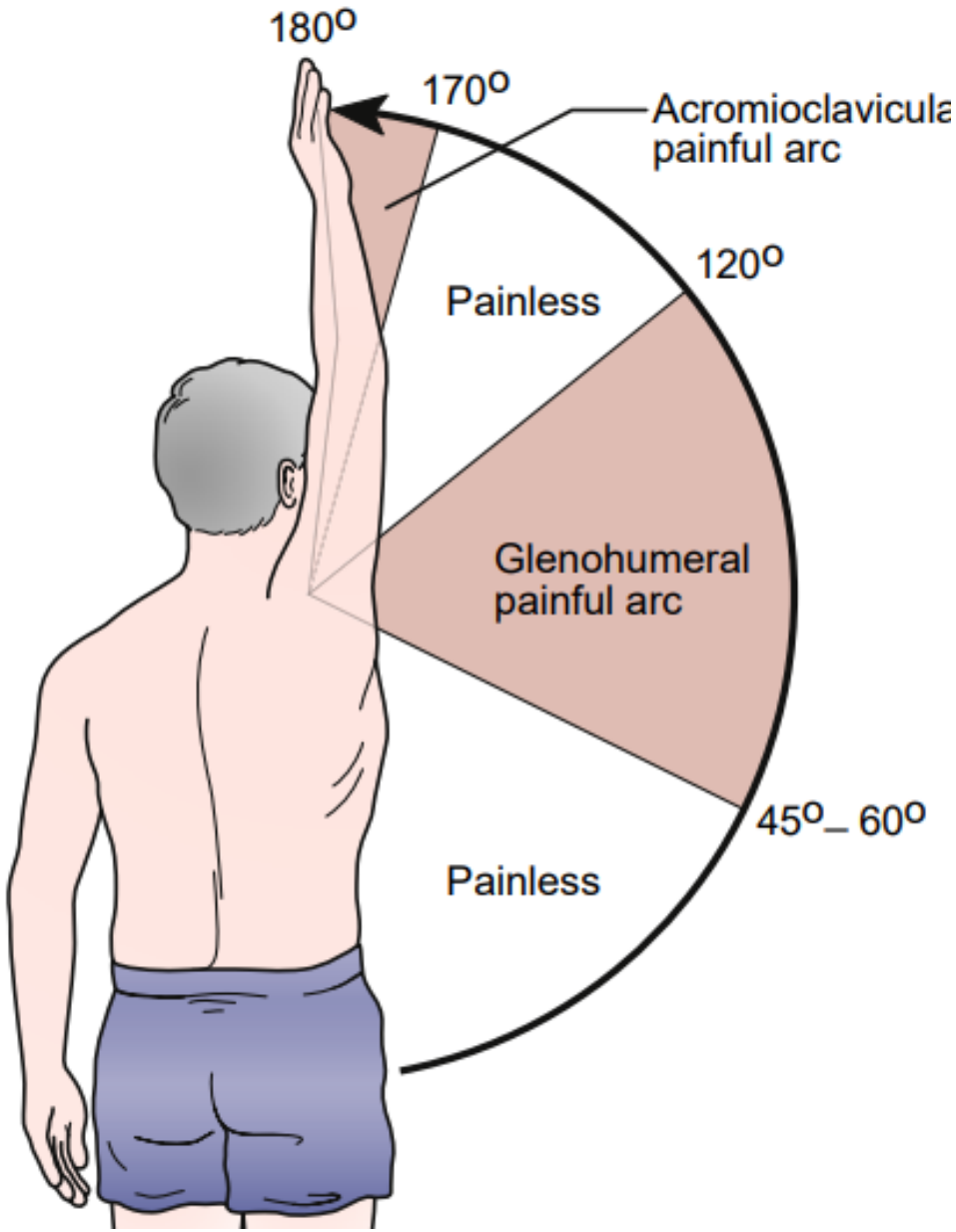
ROM

MRC

Special Tests

Neural and vascular test

Shoulder Impingement: Painful Arc test



Observation

Palpation

ROM

MRC

Special Tests

Neural and vascular test

Rotator cuff pathology

Subscapularis tendinopathies

Lift off test



Supraspinatus and infraspinatus tear

External Rotation lag sign



Observation

Palpation

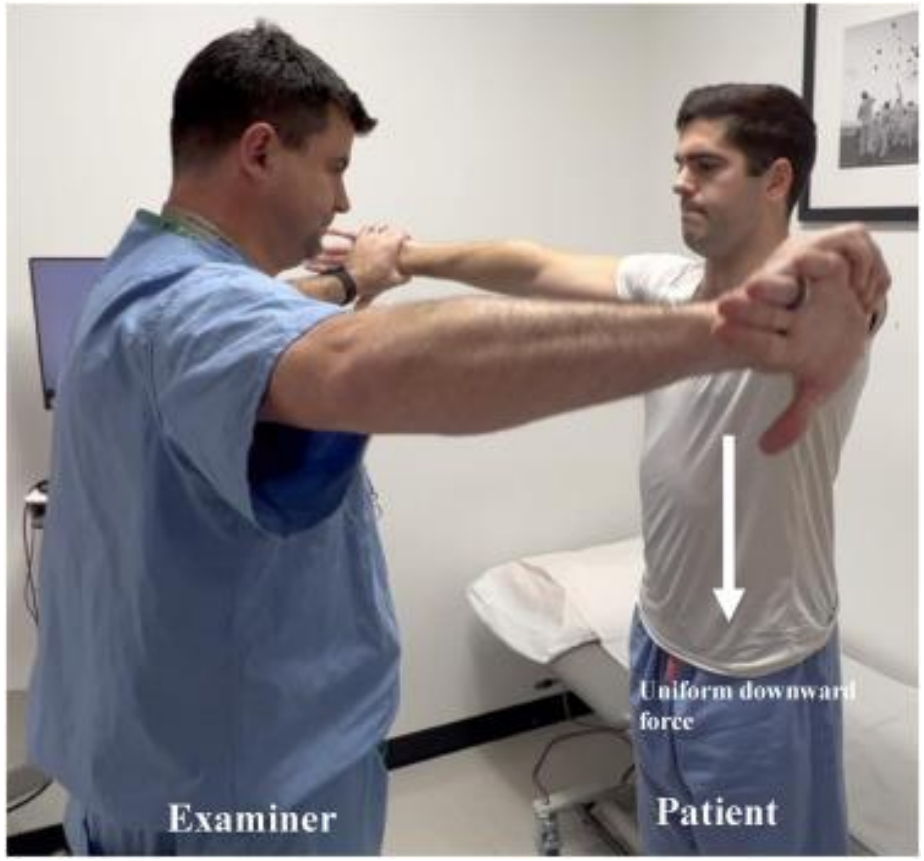
ROM

MRC

Special Tests

Neural and vascular test

Supraspinatus tear or tendinopathy close and open can (Jobe) test



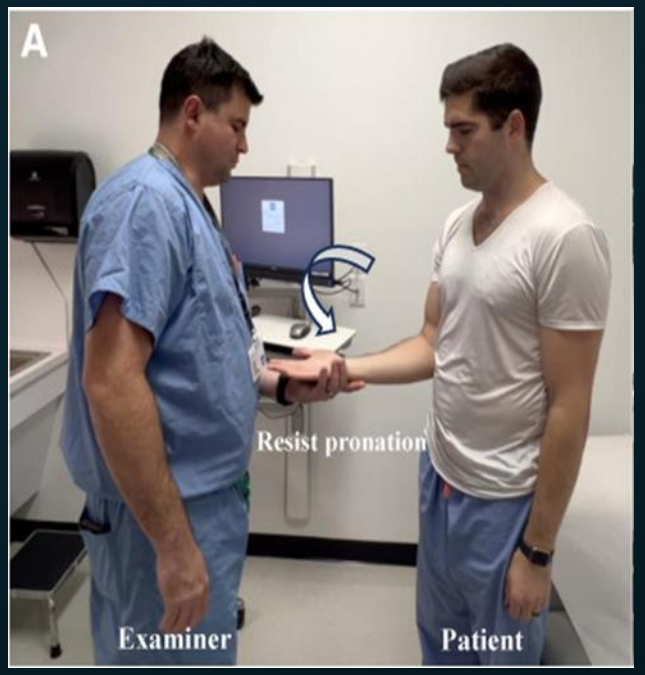


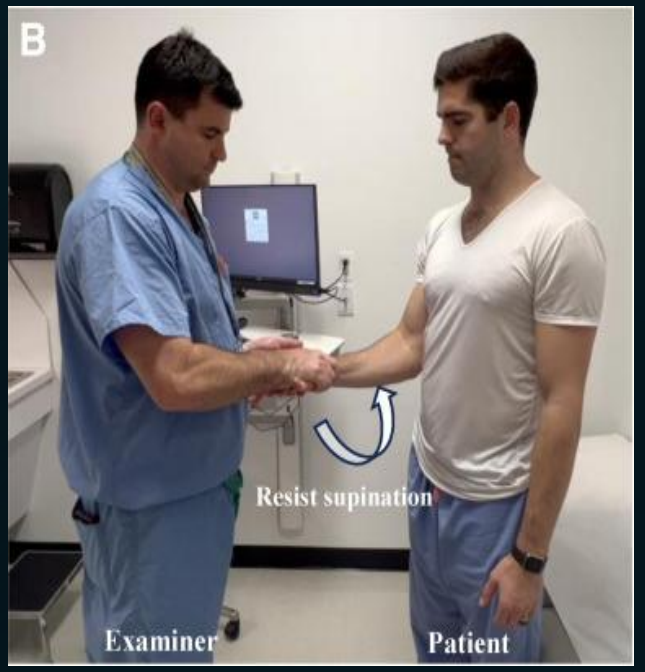
Biceps tendinopathy test:

Bilateral Speeds test and Yergason's and palpation of LHB tendon







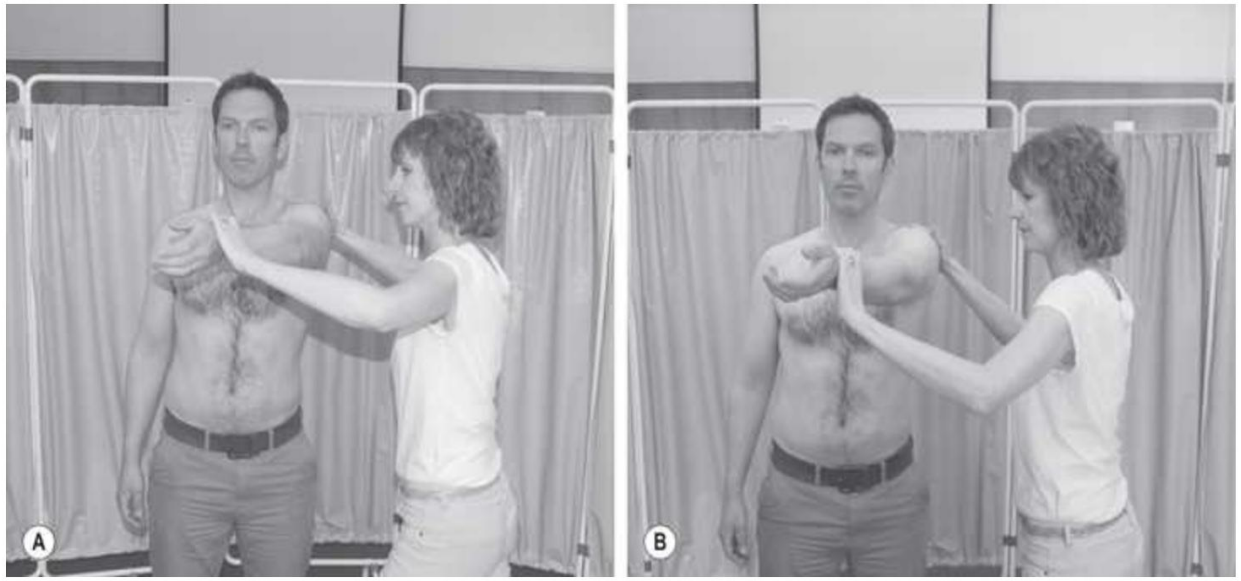


Anterior and posterior labral tests

- Biceps Load Test II



Obrien test (active compression)



Observation

Palpation

ROM

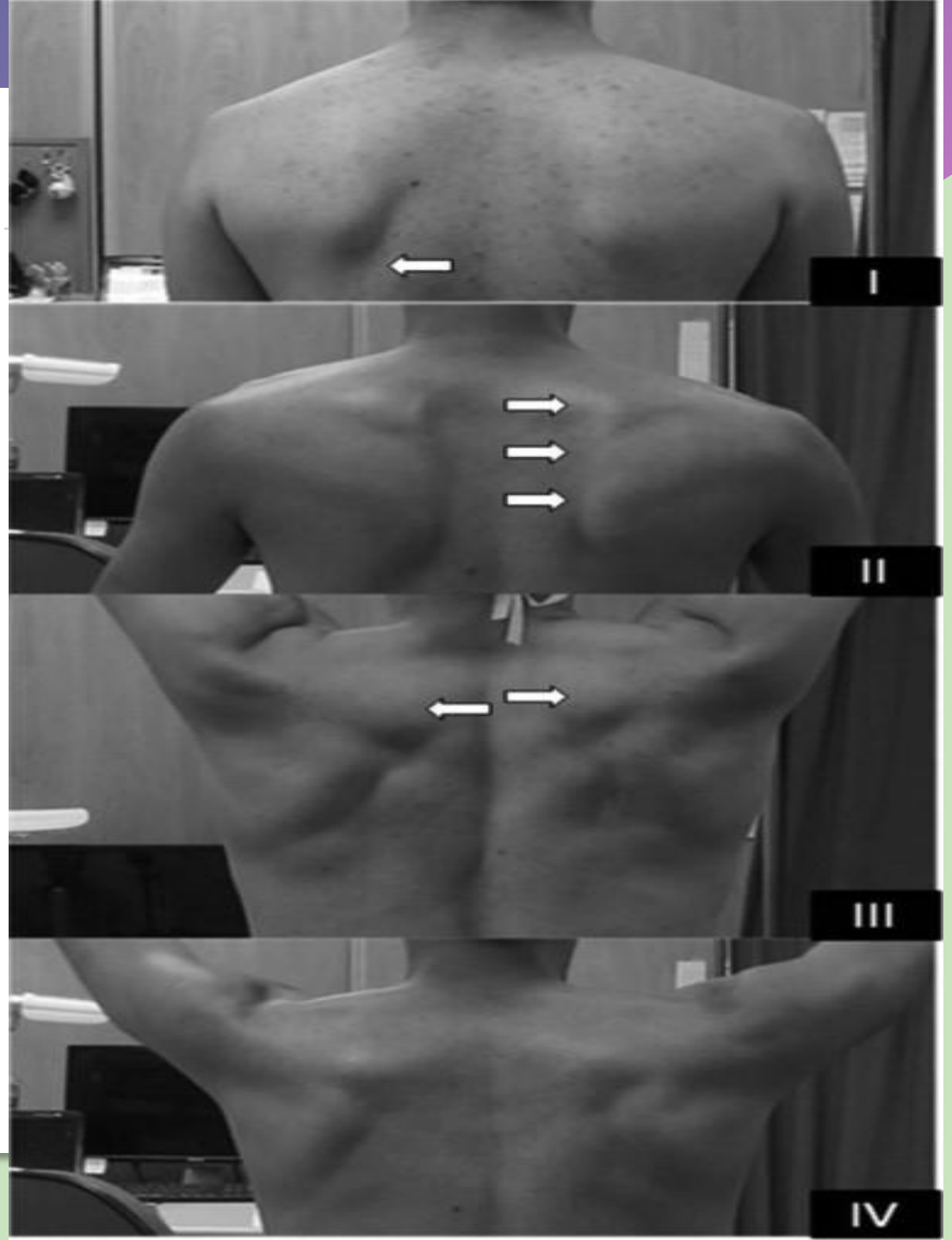
MRC

Special Tests

Neural and vascular test

Scapular Dyskinesis test



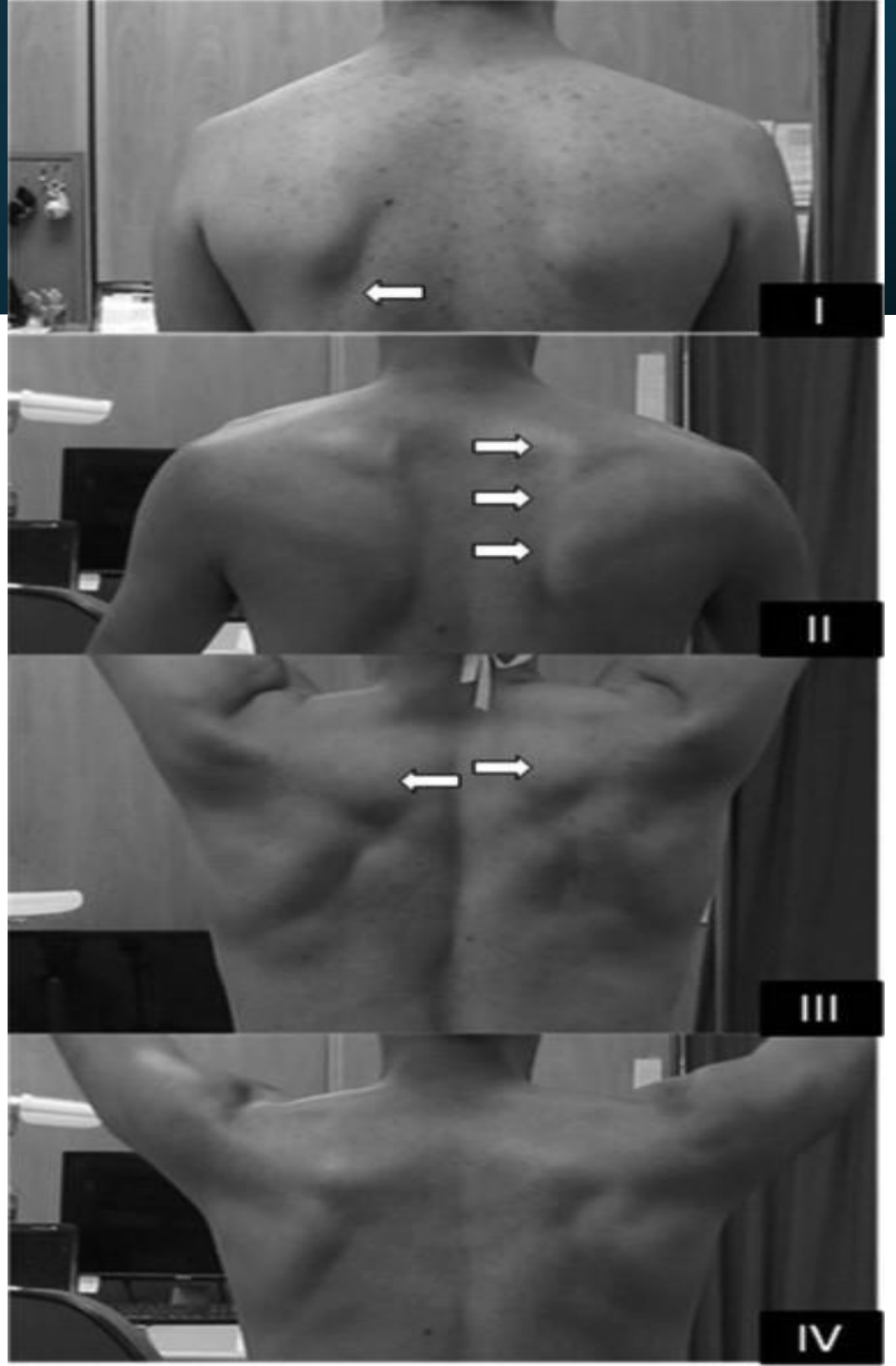


Scapular Dyskinesia Mechanism Associated Effects

- Lesser scapular upward rotation and medial boarder elevation: Inadequate serratus anterior activation
- Elevation of inferior pole of the scapula: Weakness of lower and middle trapezius.
- Scapular protraction: Wekness of middle trapezius and rhomboids.
- Greater clavicular elevation (shoulders shrugged) : Excess upper trapezius activation
- Greater scapular medial rotation and anterior tilt: Pectoralis minor tightness

Descriptions of scapular dyskinesis patterns

Pattern	Descriptions
Pattern I	Inferior medial angle of the scapula is displaced posteriorly from the posterior thorax, prominent during dynamic observation and palpation
Pattern II	Entire medial border of the scapula is displaced posteriorly from the posterior thorax, prominent during dynamic observation and palpation
Pattern III	Early scapular elevation or excessive/insufficient scapular upward rotation (dysrythmia) during dynamic observation and palpation compared to asymptomatic side
Pattern IV (Normal)	•1No evidence of posterior displacement in medial border/inferior angle of the scapula and excessive/insufficient scapular movement •2Minimal motion during the initial 30–60° of scapulohumeral elevation, then smoothly and continuously rotate upward and downward during humeral elevation and lowering, respectively
Mixed patterns	Mixed condition of above abnormal patterns (1. pattern I + II 2. pattern II + III 3. pattern I + III 4. pattern I + II + III)



Observation

Palpation

ROM

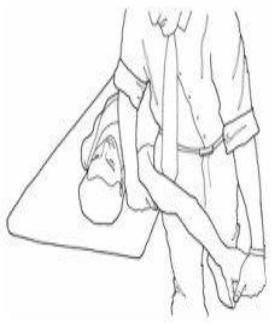
MRC

Special Tests

Neural and vascular test

Neurological testing

Neuro integrity testing for neurological symptoms



ULNT$_{MEDIAN\,(1)}$



ULNT$_{MEDIAN\,(2a)}$



ULNT$_{RADIAL\,(2b)}$



ULNT$_{ULNAR\,(3)}$

Neurodynamic testing

Vascular testing

- Adeson's
- Roos Test

Observation

Palpation

ROM

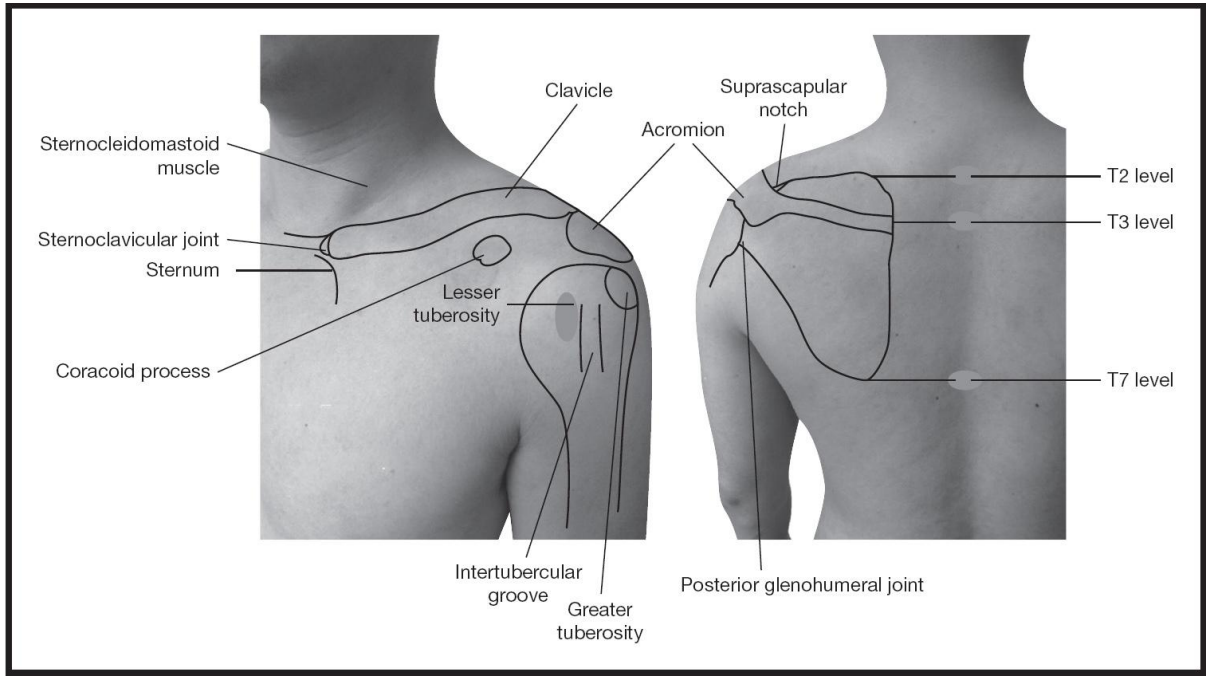
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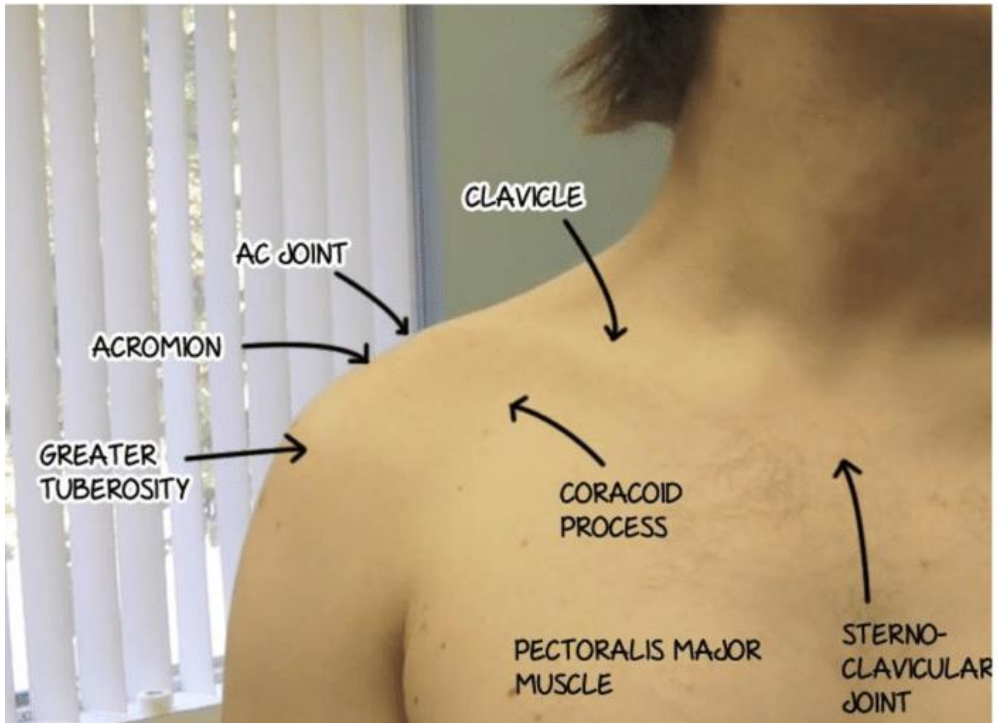
Special Tests

Neural and vascular test

Palpation

- Joint line
- Soft tissue
- Bursae
- Tendons
- Muscles
- Ligaments
- Bony prominences






Occipital border
Upper trapezius
Levator scapulae insertion
Levator scapulae
Neck extensors
Supraspinatus
Infraspinatus
Deltoid

Questions?